Zeta Register over $\mathbb{F}_{13}$:
What Every Equation and Every Layer Does for the System

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Abstract

This article provides a detailed analysis of all layers, equations, and constructions from the document "Zeta Register over F13: SymPy Verified." Each section is derived exclusively from the source text, presents the exact equations in their original form, and explains what each specific equation or layer **does for the Zeta system**: without speculation, without shortening, and without adding information not explicitly stated in the source document.

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1 Layer 1: $\mathbb{F}_{13}^3$: Radix-13 Encoding of the Input Byte

Mathematical object: Three-dimensional vector space over the finite field $\mathbb{F}_{13}$.

Number of states: $13^3 = 2197$.

Key equation:

$$b = a_0 + 13a_1 + 13^2 a_2 \quad \Longrightarrow \quad \mathbf{a} = (a_0, a_1, a_2) \in \mathbb{F}_{13}^3 \tag{1}$$

What this layer does for the Zeta system:

This layer solves the fundamental problem of representing input data within the system. Since conventional computer systems operate with bytes $b \in \{0, 1, \ldots, 255\}$, the Zeta Register requires a canonical way to translate these bytes into its algebraic structure. The equation $b = a_0 + 13a_1 + 13^2 a_2$ performs a **Radix-13 decomposition**, which uniquely splits every byte into a triple of coefficients (a_0, a_1, a_2), where each coefficient lies in the field $\mathbb{F}_{13}$. This means that the entire range of 256 byte values is encoded into a carrier space with 2197 possible states, providing a **native space for arithmetic operations** without the need to recode into other forms. The triple $\mathbf{a}$ is the invariant of this layer: every byte has exactly one representation, and this representation is directly usable in the cubic, projective, and spectral operations of the higher layers.

2 Layer 2: $T_3 \in SL(3, \mathbb{F}_{13})$: Orbital Dynamics of the Matrix

Mathematical object: Order-3 matrix in the special linear group over $\mathbb{F}_{13}$.

Number of orbital positions: 168.

Key equations:

$$\det(T_3) = 1, \qquad \mathrm{Tr}(T_3) = 1 \tag{2}$$

$$\mathbf{s}_n = T_3^n \, \mathbf{s}_0, \qquad n \in \mathbb{Z}/168\mathbb{Z} \tag{3}$$

What this layer does for the Zeta system:

T_3 is the **canonical operator** of the entire system. The conditions $\det = 1$ and $\mathrm{Tr} = 1$ ensure that the matrix belongs to the group $SL(3, \mathbb{F}_{13})$, meaning it preserves volume (the determinant is unity) and has a fixed trace. The equation $\mathbf{s}_n = T_3^n \mathbf{s}_0$ defines the **orbital dynamics**: every state $\mathbf{s}_0$ evolves in time n by repeated multiplication with the matrix T_3. Because $n \in \mathbb{Z}/168\mathbb{Z}$, this dynamics is **periodic with period 168**. This means that after 168 steps the system returns exactly to its initial state. This period of 168 generates **exact clocks without drift**: no numerical error accumulates, because everything proceeds in a finite field. The orbit of order 168 is the foundation for the Wilson Register, CRT parallelism, and synchronization of metallic elements.

3 Layer 3: P_1, P_{23}: Spectral Decomposition

Mathematical object: Projector matrices.

Number of states: $13 + 169 = 182$ (split between two channels).

Key equations:

$$\mathbf{s} = P_1 \mathbf{s} + P_{23} \mathbf{s} \tag{4}$$

$$P_1^2 = P_1, \qquad P_{23}^2 = P_{23}, \qquad P_1 P_{23} = 0 \tag{5}$$

What this layer does for the Zeta system:

This layer performs a **spectral decomposition** of the state $\mathbf{s}$ into two orthogonal subspaces. The projector P_1 decomposes the state into the **coarse channel** with dimension 1 and 13 states, while P_{23} decomposes into the **detail channel** with dimension 2 and 169 states. The conditions $P_1^2 = P_1$ and $P_{23}^2 = P_{23}$ ensure that these are genuine projectors: repeated application yields the

same result. The condition $P_1 P_{23} = 0$ ensures **orthogonality**: the coarse and detail channels are disjoint. For the Zeta system this means that information is split into two independent streams: one with low resolution (13 states) and one with high resolution (169 states). This decomposition enables **selective processing**: for example, fast verification through the coarse channel and detailed analysis through the detail channel.

4 Layer 4: $\mathbb{F}_{169} = \mathbb{F}_{13}[\theta]/(\theta^2 - 2)$: Quadratic Field Extension

Mathematical object: Quadratic extension of $\mathbb{F}_{13}$ by adjoining the element θ.

Number of states: 169.

Key equations:

$$\theta^2 = 2 \tag{6}$$

$$\lambda_{2,3} = 10 \pm 7\theta \tag{7}$$

What this layer does for the Zeta system:

This layer extends the base field $\mathbb{F}_{13}$ to $\mathbb{F}_{169}$, thereby enabling work with **complex structures in a finite field**. The equation $\theta^2 = 2$ defines θ as a square root of 2 in the field $\mathbb{F}_{13}$. Because 2 is a quadratic non-residue in $\mathbb{F}_{13}$, this extension has degree 2 and creates a field with 169 elements. The eigenvalues $\lambda_{2,3} = 10 \pm 7\theta$ are **conjugate elements** in this extended field. For the Zeta system this means that spectral analysis can work with eigenvalues that are not contained in the base field, thereby unlocking **diagonalization in a larger space**. This extension is necessary for the complete spectral decomposition of the matrix T_3 and for the construction of metallic elements in $\mathbb{F}_{169}$.

5 Layer 5: Norm Torus $T = \ker(N)$: Elements of Norm 1

Mathematical object: Kernel of the norm map.

Number of states: 14.

Key equation:

$$N(a, b) = a^2 - 2b^2 = 1 \tag{8}$$

What this layer does for the Zeta system:

The norm torus T consists of elements $(a, b) \in \mathbb{F}_{169}$ for which $a^2 - 2b^2 = 1$ holds. This equation is a **norm condition** in the quadratic extension $\mathbb{F}_{169}/\mathbb{F}_{13}$. For the Zeta system these 14 elements represent a **special subset with the algebraic structure of a torus**. Elements of norm 1 are closed under multiplication and form a cyclic subgroup. This layer provides a **controlled space for cryptographic operations**: working with norm-1 elements ensures certain invariants that are useful in key construction and identity verification within the extended field.

6 Layer 6: Cubic_{13}: Cubic Arithmetic

Mathematical object: Polynomial ring over $\mathbb{F}_{13}$ modulo a cubic polynomial.

Number of states: 2197.

Key equation:

$$\eta^3 = \eta^2 + \eta + 1 \tag{9}$$

What this layer does for the Zeta system:

This layer defines **cubic arithmetic** in the ring $\mathbb{F}_{13}[\eta]/(\eta^3 - \eta^2 - \eta - 1)$. The reduction rule $\eta^3 = \eta^2 + \eta + 1$ allows multiplication of states represented as cubic polynomials without the result exceeding degree 2. For the Zeta system this means that the product of vectors $\mathbf{a} \cdot \mathbf{b}$ in Cubic_{13} is a **fully defined algebraic operation** that requires no approximation or rounding. Every product is reduced exactly according to the given equation, preserving closure in the

space of 2197 states. This cubic structure is directly linked to the Radix-13 encoding: the triple (a_0, a_1, a_2) is interpreted as the coefficients of the polynomial $a_0 + a_1\eta + a_2\eta^2$.

7 Layer 7: $\mathbb{Z}/168\mathbb{Z}$: CRT Decomposition of the Time Axis

Mathematical object: Cyclic group of order 168.

Number of states: 168.

Key equations:

$$168 = 8 \cdot 3 \cdot 7 \tag{10}$$

$$n \mapsto (n \bmod 8,\ n \bmod 3,\ n \bmod 7) \tag{11}$$

$$n = 105(n \bmod 8) + 112(n \bmod 3) + 120(n \bmod 7) \pmod{168} \tag{12}$$

What this layer does for the Zeta system:

This layer uses the **Chinese Remainder Theorem (CRT)** to decompose the time parameter n into three independent parallel channels. The equation $168 = 8 \cdot 3 \cdot 7$ expresses the factorization of the period into pairwise coprime components. The map $n \mapsto (n \bmod 8, n \bmod 3, n \bmod 7)$ splits the computation into three independent subproblems that can run concurrently. The reconstruction formula $n = 105(n \bmod 8) + 112(n \bmod 3) + 120(n \bmod 7) \pmod{168}$ allows **exact reconstruction** of the original time from the three parallel results. For the Zeta system this means that orbital dynamics can be **parallelized without loss of information**: each channel works with a smaller period (8, 3, or 7), which speeds up computation and enables distributed implementation.

8 Layer 8: $\mathbb{Z}/13^k\mathbb{Z}$: Hensel Lifting

Mathematical object: Ring of integers modulo 13^k.

Number of states: 13^k.

Key equation:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \tag{13}$$

What this layer does for the Zeta system:

Hensel lifting allows **iterative lifting of polynomial solutions** from the field $\mathbb{F}_{13}$ to the rings $\mathbb{Z}/13^k\mathbb{Z}$. The equation $x_{k+1} = x_k - f(x_k)/f'(x_k)$ is Hensel's lemma in action: step by step the precision of the solution is increased by adding higher powers of the prime 13. For the Zeta system this means that results verified modulo 13 can be **exactly extended** to higher powers of the modulus without loss of consistency. This enables the transition from the coarse field $\mathbb{F}_{13}$ to finer structures where more information about the arithmetic structure is available.

9 Layer 9: Witt Vectors $\mathbb{W}_2(\mathbb{F}_{13})$: p-adic Encoding

Mathematical object: Witt vectors of length 2 over $\mathbb{F}_{13}$.

Number of states: 169.

Key equations:

$$[a] = a^{13} \pmod{169} \tag{14}$$

$$(a_0, a_1) \in \mathbb{F}_{13}^2 \longleftrightarrow \mathbb{Z}/169\mathbb{Z} \quad \text{via the ghost map} \tag{15}$$

What this layer does for the Zeta system:

Witt vectors provide a **canonical bridge** between pairs of elements from $\mathbb{F}_{13}$ and elements of the ring $\mathbb{Z}/169\mathbb{Z}$. The Teichmüller representative $[a] = a^{13} \pmod{169}$ maps an element $a \in \mathbb{F}_{13}$ into the ring modulo 169 such that it preserves its algebraic properties: specifically $[a]^{13} \equiv [a]$

(mod 169). The ghost map then defines a bijection between pairs (a_0, a_1) and elements of $\mathbb{Z}/169\mathbb{Z}$. For the Zeta system this means that **p-adic arithmetic is exactly encoded** and the structure $\mathbb{Z}/169\mathbb{Z}$ is accessible through the vector representation in $\mathbb{F}_{13}^2$.

10 Layer 10: $PG(2, \mathbb{F}_{13})$: Projective Geometry

Mathematical object: Projective plane over $\mathbb{F}_{13}$.

Number of states: 183 points, 183 lines, 2562 flags.

Key equation:

$$\frac{13^3 - 1}{13 - 1} = 183 \tag{16}$$

What this layer does for the Zeta system:

The projective plane $PG(2, \mathbb{F}_{13})$ provides a **geometric interpretation of the system's states**. The equation $(13^3 - 1)/(13 - 1) = 183$ gives the number of points in the projective plane: each point corresponds to a one-dimensional subspace in $\mathbb{F}_{13}^3$. With 183 lines and 2562 flags (incident point-line pairs), this layer creates a **holographic structure** where each point lies on 14 lines and each line passes through 14 points. For the Zeta system this means that states can be **geometrically projected** with exact incidence: relationships between states are representable as geometric relationships in the projective plane.

11 Layer 11: Rank Decompositions: Tree Addressing

Mathematical object: Decomposition of the 2197-state space by rank.

Number of states: $2197 = 13 + 468 + 1716$.

What this layer does for the Zeta system:

Rank decomposition splits the space of 2197 states into three subsets according to the rank of the matrix representing the state: 13 states of rank 1, 468 states of rank 2, and 1716 states of rank 3. For the Zeta system this means that **every state is assigned a hierarchical level** according to its algebraic complexity. Rank-1 states are the simplest and most degenerate, while rank-3 states are full and carry the maximum amount of information. This decomposition enables **tree addressing**: efficient searching and classification of states by their algebraic rank.

12 Layer 12: Non-Squares in $\mathbb{F}_{13}^*$: Quadratic Residues

Mathematical object: Set of quadratic non-residues in the multiplicative group $\mathbb{F}_{13}^*$.

Number of states: 6.

Key equation:

$$\{2, 5, 6, 7, 8, 11\} \tag{17}$$

What this layer does for the Zeta system:

Quadratic non-residues in $\mathbb{F}_{13}^*$ are those elements that are not squares of any element in the field. The set $\{2, 5, 6, 7, 8, 11\}$ forms exactly half of the multiplicative group (which has 12 elements). For the Zeta system this layer provides a **binary classification of elements**: every non-zero element is either a square or a non-square. This is the foundation for quadratic extensions (such as $\mathbb{F}_{169}$), for the construction of the Legendre symbol, and for generating structures that require distinguishing between these two classes.

13 Layer 13: 24-Cell (Icositetrachoron): Coset Decomposition

Mathematical object: Regular 4-dimensional polytope with 24 vertices.

Number of states: 24.

Key equation:

$$C_0, C_1, C_2 \quad \text{with chirality } \pm 1 \tag{18}$$

What this layer does for the Zeta system:

The 24-cell is a special regular polytope in 4 dimensions whose 24 vertices are partitioned into three cosets C_0, C_1, C_2 with an assigned chiral sign ± 1. For the Zeta system this layer provides a **geometric model with internal symmetry**: the partition into cosets corresponds to a certain algebraic structure and the chiral sign allows distinguishing between "left" and "right" variants. This structure is usable for encoding states with orientation and for constructing dualities within the system.

14 Layer 14: Geometric Bridges: Translation of Structures

Mathematical object: Bijections between different representations.

Key equation:

$$24 \longleftrightarrow \mathbb{F}_{13}^* \times C_2 \tag{19}$$

What this layer does for the Zeta system:

This layer defines **explicit bijections** between geometric and algebraic structures. Specifically, the equation $24 \longleftrightarrow \mathbb{F}_{13}^* \times C_2$ expresses that the 24 vertices of the 24-cell are in bijection with the Cartesian product of the multiplicative group $\mathbb{F}_{13}^*$ (which has 12 elements) and the cyclic group C_2 (which has 2 elements), giving $12 \times 2 = 24$. For the Zeta system this means that **geometric and algebraic representations are mutually translatable**: operations performed in one representation can be exactly converted into the other without loss of information.

15 Layer 15: Correct Circulant: 24-Hour Mechanism

Mathematical object: Circulant matrix representing a periodic structure.

Number of states: 12.

Key equation:

$$T_3^{\text{orbit}} = 7I + 2C + 6C^2 \tag{20}$$

What this layer does for the Zeta system:

The correct circulant expresses the orbital structure of the matrix T_3 in a circulant basis, where C is the basic circulant matrix (shift by one position) and I is the identity. The equation $T_3^{\text{orbit}} = 7I + 2C + 6C^2$ states that the orbital dynamics is a **combination of the identity and two shifts** with coefficients 7, 2, and 6 in the field $\mathbb{F}_{13}$. For the Zeta system this means that **periodic evolution has a simple circulant representation**, which is efficiently implementable and analyzable using Fourier transforms.

16 Layer 16: $\mathfrak{sl}(3, \mathbb{F}_{13})$: Lie Algebra

Mathematical object: Special linear Lie algebra over $\mathbb{F}_{13}$.

Number of states: 13^8.

Key equation:

$$\text{Jacobi identity for } 8^3 = 512 \text{ triples} \tag{21}$$

What this layer does for the Zeta system:

The Lie algebra $\mathfrak{sl}(3, \mathbb{F}_{13})$ is the algebraic structure associated with the Lie group $SL(3, \mathbb{F}_{13})$. It has dimension 8, giving 13^8 elements. The Jacobi identity is the fundamental axiom of a Lie algebra, which must hold for all triples of elements: in this case for $8^3 = 512$ triples of generators. For the Zeta system this layer provides an **infinitesimal view of the symmetries** of the group $SL(3, \mathbb{F}_{13})$. It allows the study of local properties of transformations and the construction of exponential maps between the algebra and the group.

17 Layer 17: Wilson Register: Periodic Signature

Mathematical object: Trace of powers of the matrix T_3.

Number of states: 168.

Key equation:

$$W(n) = \mathrm{Tr}(T_3^n) \pmod{13} \tag{22}$$

What this layer does for the Zeta system:

The Wilson Register is a **168-value periodic signature** of the system's state. The equation $W(n) = \mathrm{Tr}(T_3^n) \pmod{13}$ defines for every time step n a value that is the trace of the n-th power of the canonical matrix. Because the period is 168, this signature is **fully deterministic and periodic**. For the Zeta system the Wilson Register serves as a **hashing function**: a compact representation of the state that depends on the entire orbital history. It is usable for integrity verification, pseudorandom sequence generation, and as input for modular forms.

18 Layer 18: Fermion Module: Dirac Operator

Mathematical object: Dirac operator over a finite field.

Number of states: 13^6.

Key equations:

$$D_f = \sigma_1 \otimes D_W \quad \text{over } \mathbb{F}_{13}^6 \tag{23}$$

$$\det(D_f + mI) = 0 \quad \text{for } m = 5, 8 \tag{24}$$

What this layer does for the Zeta system:

The fermion module extends the system to a **6-dimensional space** $\mathbb{F}_{13}^6$, where the Dirac operator D_f acts as the tensor product of the Pauli matrix σ_1 and the Wilson operator D_W. The chiral projectors $P_\pm$ and the parity operator $\gamma_5 = \sigma_3 \otimes G$ split states into left and right components. The singular masses $m = 5$ and $m = 8$ are those values for which the operator $D_f + mI$ has zero determinant: meaning that **non-trivial kernels (zero modes) exist**. For the Zeta system this represents a **finite-field analogue of the chiral anomaly**: a situation where symmetry is not preserved due to the existence of specific mass parameters.

19 Layer 19: NTT: Fast Fourier Transform

Mathematical object: Number Theoretic Transform.

Length: 12.

Key equations:

$$\omega = 2 \quad \text{(primitive root)} \tag{25}$$

$$\det(W) = 5 \tag{26}$$

$$W_{k,n}^{-1} = 12 \cdot 2^{-kn} \pmod{13} \tag{27}$$

What this layer does for the Zeta system:

The NTT (Number Theoretic Transform) is a **Fourier transform over a finite field**. The primitive root $\omega = 2$ in $\mathbb{F}_{13}$ has order 12, which enables a transform of length 12. The equation $\det(W) = 5$ gives the determinant of the transform matrix. The inverse NTT is defined as $W_{k,n}^{-1} = 12 \cdot 2^{-kn} \pmod{13}$. For the Zeta system the NTT provides **spectral analysis in a finite field**: it enables transition between the time and frequency domains without using real numbers and without rounding errors. The Cooley-Tukey butterfly structure ensures efficient implementation in $O(N \log N)$ time.

20 Layer 20: Tits Tree: 14-Regular Tree

Mathematical object: Bruhat-Tits tree for the group $SL(2)$ over a local field.

Structure: 14-regular tree with finite residue.

Key equation:

$$\text{Residue } PG(1, \mathbb{F}_{13}) \tag{28}$$

What this layer does for the Zeta system:

The Tits Tree (Bruhat-Tits tree) is an **infinite 14-regular tree**, where every vertex has degree 14. The residue at each vertex is the projective line $PG(1, \mathbb{F}_{13})$. For the Zeta system this layer provides a **hierarchical structure for classification and organization of states**. The tree structure allows gradual refinement or coarsening of representations, where each level of the tree corresponds to a certain granularity of information. It is an abstract framework for **addressing and navigation** in the state space of the system.

21 Layer 21: $\Gamma(13)$ Modular Forms: Weight Functions on Cusps

Mathematical object: Modular forms for the principal congruence subgroup of level 13.

Number of cusps: 168.

Key equations:

$$f_2(n) = W(n) \tag{29}$$

$$T_{13} f = f \tag{30}$$

What this layer does for the Zeta system:

Modular forms for $\Gamma(13)$ are functions on the upper half-plane with a specific transformation property. The equation $f_2(n) = W(n)$ identifies the weight function f_2 with the Wilson Register $W(n)$. The condition $T_{13} f = f$ states that this function is an **eigenfunction of the Hecke operator** T_{13} with eigenvalue 1. For the Zeta system this layer provides a **deep number-theoretic structure**: the Wilson Register is not merely a random periodic sequence, but has an interpretation as a modular form, which opens possibilities for using deep results from number theory (for example, from the Langlands philosophy) to analyze the properties of the system.

22 Layer 22: MERA Pyramid: Tensor Compression

Mathematical object: Multi-scale entanglement renormalization ansatz.

Number of levels: 6.

Key equations:

$$168 \xrightarrow{G} 84 \xrightarrow{P_1} 42 \xrightarrow{W_3} 14 \xrightarrow{W_7} 7 \xrightarrow{W_8} 1 \tag{31}$$

$$S_{\text{ent}} = \log_{13}(\text{rank of reduced matrix}) \tag{32}$$

What this layer does for the Zeta system:

The MERA pyramid performs **progressive coarse-graining**: refinement of the state space from 168 through 84, 42, 14, 7 down to 1 state. Each step uses a specific operator (G, P_1, W_3, W_7, W_8). The entanglement entropy $S_{\text{ent}} = \log_{13}(\text{rank of reduced matrix})$ measures how much information is lost or preserved at each coarse-graining step. For the Zeta system this layer provides **hierarchical compression**: from 3 bits of information at the finest level down to 0 bits after complete coarse-graining. It allows analyzing the system at different scales and studying how local properties manifest at the global level.

23 Layer 23: Bulk/Boundary: Holographic Projection

Mathematical object: Holographic duality between bulk and boundary.

Number of states: 2562 bulk chambers / 183 boundary points.

Key equation:

$$\frac{2562}{183} = 14 \tag{33}$$

What this layer does for the Zeta system:

The holographic projection maps **2562 bulk chambers** (flags in the projective plane) onto **183 boundary points** (points of the projective plane). The ratio $2562/183 = 14$ states that each boundary point "encodes" information from 14 bulk chambers. For the Zeta system this means that **high-dimensional information is compressible into a lower-dimensional boundary** according to an exact ratio. This layer enables efficient storage and transmission of information: instead of working with 2562 states in the bulk, it suffices to work with 183 states on the boundary, while the original information is reconstructible.

24 Layer 24: Metallic Elements: Generators of the Extended Field

Mathematical object: Algebraic elements with special power properties.

Number of symbols: 4.

Key equations:

$$\varphi^2 = \varphi + 1, \qquad \gamma^{168} = 1 \tag{34}$$

$$\varphi^{28} = 1, \qquad \beta^4 = 1, \qquad \delta^{28} = 1 \tag{35}$$

What this layer does for the Zeta system:

Metallic elements are **special algebraic elements** in extended fields with specific periods. The golden ratio φ has in $\mathbb{F}_{169}$ the same definition as in the real numbers ($\varphi^2 = \varphi + 1$), but is a cyclic element of order 28. The element γ has order 168, β has order 4, and δ has order 28. For the Zeta system these elements provide:

1. **Pseudorandomness:** Powers φ^k generate deterministic sequences with good mixing properties: each power produces a unique permutation of 28 states.
2. **Subgroup hierarchy:** The structure $C_{168} \supset C_{28} \supset C_4 \supset C_2 \supset C_1$ gives a natural hierarchy of security classes.
3. **Synchronization:** Each metallic element defines its own periodic timer. The common period of all timers is $\mathrm{lcm}(168, 28, 4) = 168$, which means that distributed processes resynchronize after 168 steps.

25 Complete Data Flow of the System: 13-Step Pipeline

The entire Zeta Register forms a **unified exact pipeline**, where the output of one layer is the exact input of the next. Below is a detailed breakdown of each step.

25.1 Step 1: Input Byte

$$b \in \{0, 1, \ldots, 255\} \tag{36}$$

The system accepts a standard byte as input. This byte is an integer from 0 to 255.

25.2 Step 2: Radix-13 Decomposition

$$b = a_0 + 13a_1 + 13^2 a_2 \quad \Longrightarrow \quad \mathbf{a} = (a_0, a_1, a_2) \in \mathbb{F}_{13}^3 \tag{37}$$

The byte is uniquely decomposed into a triple of coefficients in the field $\mathbb{F}_{13}$. Every byte is encoded as a triple in a carrier with 2197 states. This step ensures that input data is in the system's native format.

25.3 Step 3: Cubic Arithmetic

$$\mathbf{a} \cdot \mathbf{b} \text{ in } \mathrm{Cubic}_{13} = \mathbb{F}_{13}[\eta]/(\eta^3 - \eta^2 - \eta - 1) \tag{38}$$

Multiplication of states proceeds in the cubic ring with the reduction rule $\eta^3 = \eta^2 + \eta + 1$. This step enables algebraic operations over states without exceeding the dimension of the space.

25.4 Step 4: Spectral Decomposition

$$\mathbf{s} = P_1\mathbf{s} + P_{23}\mathbf{s} \tag{39}$$

The state $\mathbf{s}$ is decomposed into the coarse channel $P_1\mathbf{s}$ (dimension 1, 13 states) and the detail channel $P_{23}\mathbf{s}$ (dimension 2, 169 states). This step separates information by resolution.

25.5 Step 5: Diagonalization over $\mathbb{F}_{169}$

$$\mathbf{s} = Q_1\mathbf{s} + Q_2\mathbf{s} + Q_3\mathbf{s} \tag{40}$$

The state is further decomposed according to eigenvalues: Q_1 corresponds to $\lambda_1 = 7$ in $\mathbb{F}_{13}$, while Q_2, Q_3 correspond to the conjugate values $\lambda_{2,3} = 10 \pm 7\theta$ in $\mathbb{F}_{169}$. This step provides a complete spectral description of the state.

25.6 Step 6: Orbital Dynamics

$$\mathbf{s}_n = T_3^n\, \mathbf{s}_0, \qquad n \in \mathbb{Z}/168\mathbb{Z} \tag{41}$$

The state evolves according to powers of the canonical matrix T_3. The period of 168 generates exact clocks without drift: temporal evolution is deterministic and periodic.

25.7 Step 7: CRT Parallelism

$$n \mapsto (n \bmod 8,\ n \bmod 3,\ n \bmod 7) \tag{42}$$

The time parameter n is decomposed into three independent parallel channels. Reconstruction proceeds according to the formula:

$$n = 105(n \bmod 8) + 112(n \bmod 3) + 120(n \bmod 7) \pmod{168} \tag{43}$$

This step enables parallel computation and distributed implementation.

25.8 Step 8: p-adic Lifting

- Hensel lifting of polynomial roots from $\mathbb{F}_{13}$ to $\mathbb{Z}/13^k\mathbb{Z}$ iteratively, step by step.
- Teichmüller representatives: $[a] = a^{13} \pmod{169}$ for every $a \in \mathbb{F}_{13}$.
- Witt vectors: $(a_0, a_1) \in \mathbb{F}_{13}^2 \longleftrightarrow \mathbb{Z}/169\mathbb{Z}$ via the ghost map.

This step extends arithmetic from the field $\mathbb{F}_{13}$ to p-adic rings of higher precision.

25.9 Step 9: Fourier Analysis

- NTT of length 12 with primitive root $\omega = 2$ over $\mathbb{F}_{13}$.
- Cooley-Tukey butterfly structure for efficient $O(N \log N)$ implementation.
- Inverse NTT: $W_{k,n}^{-1} = 12 \cdot 2^{-kn} \pmod{13}$.

This step provides spectral analysis in a finite field with exact inversion.

25.10 Step 10: Fermion Field and Dirac Operator

- $D_f = \sigma_1 \otimes D_W$ over $\mathbb{F}_{13}^6$.
- Chiral projectors $P_\pm$, parity operator $\gamma_5 = \sigma_3 \otimes G$.
- Singular masses $m = 5, 8$ with mixed kernels: finite-field analogue of the chiral anomaly.

This step extends the system to a fermionic structure with chiral properties.

25.11 Step 11: Geometric Projection

- States $\to PG(2, \mathbb{F}_{13})$: 183 points, 183 lines, 2562 flags.
- Holographic projection: 2562 bulk chambers $\to$ 183 boundary points (ratio 14:1).
- 24-cell: 24 vertices partitioned into three cosets C_0, C_1, C_2 with chiral duals.

This step maps algebraic states onto geometric structures.

25.12 Step 12: MERA Coarse-Graining

$$168 \xrightarrow{G} 84 \xrightarrow{P_1} 42 \xrightarrow{W_3} 14 \xrightarrow{W_7} 7 \xrightarrow{W_8} 1 \tag{44}$$

- Entanglement entropy at each level: $S_{\text{ent}} = \log_{13}(\text{rank of reduced matrix})$.
- From 3 bits of information at the finest level down to 0 bits after complete coarse-graining.

This step progressively compresses information through a hierarchical pyramid.

25.13 Step 13: Output

- Hash: 168-value Wilson register $W(n) = \text{Tr}(T_3^n) \pmod{13}$.
- Entropy: spectral information about the state after projections.
- Modular form: weight function $f_2(n) = W(n)$ on the cusps of $\Gamma(13)$.

This step produces three types of outputs: a deterministic hash, an entropic measure, and a modular representation.

Layer	Mathematical Object	States	Key Operation	Invariant

26 Overview Table of All 24 Layers

Layer	Mathematical Object	States	Key Operation	Invariant
1	$\mathbb{F}_{13}^3$	2197	Radix-13 encoding	Triple (a_0, a_1, a_2)
2	$T_3 \in SL(3, \mathbb{F}_{13})$	168	Matrix powering	$\det = 1$, $\mathrm{Tr} = 1$
3	P_1, P_{23}	$13 + 169$	Spectral projection	$P_1^2 = P_1$, $P_{23}^2 = P_{23}$, $P_1 P_{23} = 0$
4	$\mathbb{F}_{169} = \mathbb{F}_{13}[\theta]/(\theta^2 - 2)$	169	Quadratic extension	$\theta^2 = 2$, $\lambda_{2,3} = 10 \pm 7\theta$
5	Norm torus $T = \ker(N)$	14	Norm-1 elements	$N(a, b) = a^2 - 2b^2 = 1$
6	Cubic_{13}	2197	Cubic arithmetic	$\eta^3 = \eta^2 + \eta + 1$
7	$\mathbb{Z}/168\mathbb{Z}$	168	CRT decomposition	$168 = 8 \cdot 3 \cdot 7$
8	$\mathbb{Z}/13^k\mathbb{Z}$	13^k	Hensel lifting	$x_{k+1} = x_k - f(x_k)/f'(x_k)$
9	Witt vectors $\mathbb{W}_2(\mathbb{F}_{13})$	169	p-adic encoding	$[a]^{13} \equiv [a] \pmod{169}$
10	$PG(2, \mathbb{F}_{13})$	183	Projective geometry	$(13^3 - 1)/(13 - 1) = 183$
11	Rank decompositions	2197	Tree addressing	$2197 = 13 + 468 + 1716$
12	Non-squares in $\mathbb{F}_{13}^*$	6	Quadratic residues	$\{2, 5, 6, 7, 8, 11\}$
13	24-Cell (Icositetrachoron)	24	Coset decomposition	C_0, C_1, C_2 with chirality ± 1
14	Geometric bridges	Bijections	Structure translation	$24 \leftrightarrow \mathbb{F}_{13}^* \times C_2$
15	Correct circulant	12	24-hour mechanism	$T_3^{\mathrm{orbit}} = 7I + 2C + 6C^2$
16	$\mathfrak{sl}(3, \mathbb{F}_{13})$	13^8	Lie algebra	Jacobi identity for $8^3 = 512$ triples
17	Wilson Register	168	Periodic signature	$W(n) = \mathrm{Tr}(T_3^n)$
18	Fermion module	13^6	Dirac operator	$\det(D_f + mI) = 0$ for $m = 5, 8$
19	NTT	12	Fast Fourier transform	$\omega = 2$, $\det(W) = 5$
20	Tits Tree	∞ (finite residue)	14-regular tree	Residue $PG(1, \mathbb{F}_{13})$
21	$\Gamma(13)$ modular forms	168 cusps	Weight functions	$f_2(n) = W(n)$, $T_{13} f = f$
22	MERA pyramid	6 levels	Tensor compression	$S_{\mathrm{ent}} = \log_{13}(\mathrm{rank})$
23	Bulk/Boundary	2562 / 183	Holographic projection	$2562/183 = 14$
24	Metallic elements	4 symbols	Generators of $\mathbb{F}_{169}^*$	$\varphi^2 = \varphi + 1$, $\gamma^{168} = 1$

27 Conclusion: Key Advantage: Absolute Exactness

The entire Zeta Register is built on a single principle: **no approximations, no real numbers, no rounding**. Every operation in the pipeline has these properties:

- **Reversible:** Every transformation has an exact inverse modulo 13. If you know the output and the key, you can exactly reconstruct the input.
- **Deterministic:** The same input always produces the identical output on every node, at every time, on every hardware. No race conditions caused by floating-point arithmetic.
- **Finite:** No limits, no convergence, no epsilon controls. Everything is arithmetic in finite fields $\mathbb{F}_{13}$, $\mathbb{F}_{169}$ or rings $\mathbb{Z}/13^k\mathbb{Z}$.
- **Verifiable:** Every equation, every identity, and every invariant has been explicitly verified symbolically via SymPy modulo 13. No need to trust: just run the code.

If you need a system where:

- every bit has an exact algebraic interpretation and is not merely an approximation,
- every hash is fully reproducible and deterministic,
- every cipher is invertible without rounding errors and without drift,
- every simulation runs with zero numerical error and zero noise entropy,
- every geometric projection preserves incidence exactly and not asymptotically,

then the Zeta Register over $\mathbb{F}_{13}$ offers a complete, closed, and exactly verified framework: from the byte through projective geometry, Lie algebras, fermion fields all the way to modular forms: with all 24 layers interconnected via the canonical operator T_3 and its orbit of order 168. All equations are functional, all bridges are verified, and the entire system is ready for implementation as a deterministic, reversible, and exact register.